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NERVE AND MUSCLE (L H WEIMER, SECTION EDITOR)

Present Uses, Future Applications, and Technical Underpinnings of Electrical Impedance Myography

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Abstract
Purpose of Review In this article, we provide an overview of electrical impedance myography (EIM), including its technical and theoretical basis, a summary of its varied applications, and ongoing developments.
Recent Findings EIM has been used as a disease severity biomarker in a variety of disorders affecting the muscle, ranging from amyotrophic lateral sclerosis (ALS) to muscular dystrophies to disuse atrophy due to the weightlessness of space. In ALS, studies have demonstrated that major reductions in sample size in clinical trials can be achieved. Similarly, in the Duchenne muscular dystrophy, the technique tracks disease progression and is sensitive to the beneficial effect of steroids. More basic work has demonstrated that EIM can provide a non-invasive means of tracking muscle fiber size. Ongoing innovations include the development of techniques for assessing muscle contraction.
Summary EIM is gradually being adopted as a useful, practical, and convenient tool for the assessment of neuromuscular conditions.

Keywords Electrical impedance · Muscle · Myopathy · Neurogenic disease · Disuse · Injury

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Introduction

Electrical impedance myography (EIM) is a relatively new, convenient tool for bedside assessment of the muscle. Although the technique is still being developed and refined and its theoretical basis is still being fully established, it is already providing useful insights into muscle condition in a wide variety of contexts. In this review, we begin by providing a background on the basic concepts on which EIM is based, a detailed overview of its current uses and recent data, and future developmental prospects.

Electrical Bioimpedance

In "standard" electrophysiology, the main focus is the measurement of the electrical potentials generated by the nerve and the muscle. In contrast, the measuring principle of electrical bioimpedance methods, henceforth referred to simply as impedance, consists of measuring the electrical potential V generated after applying an external alternating electrical current I into the body. In its simplest form, the impedance Z can be calculated for a particular frequency measured using Ohm's law as

$$Z = \frac{V}{I}.$$

The standard notation to define the impedance is as a vector, namely $=R + jX$, where R and X are the resistance and reactance components and j represents the imaginary unit to distinguish both components. One convenient representation of impedance is plotting the reactance against the resistance. In doing so, the length of the vector is the magnitude of the impedance $|Z|$ connecting the ending point (R, X) to (0, 0).

Then applying the Pythagorean trigonometric identity, one finds the following relationship:

$$|Z| = \sqrt{R^2 + X^2} \text{ and } \theta = \tan^{-1}\frac{X}{R}$$

where θ is the phase angle described by the impedance, or equivalently,

$$R = |Z|\cos\theta \text{ and } X = |Z|\sin\theta$$

As the measurement frequency is varied from kilohertz to megahertz, the impedance defines a circular-like-shape plot. Other representations include plotting the impedance component resistance and reactance against the frequency measured.

Electrical Impedance Myography: Basic Concepts

EIM is a specific instance of the broader electrical bioimpedance field [1], in which electrical current is applied to a single muscle or group of muscles [2]. The basic underlying premise of EIM is that it can provide detailed physiological data on the composition and structure of the muscle. In general, EIM relies upon the concept that the conductivity (the ability to pass electrical current) and permittivity (ability to store electrical charge) are altered in diseased muscle(s) [3] and that these alterations are the source causing the changes measured in the generated voltage. In this simple explanation, the charge of the myofiber membrane is changed with the frequency measured so that only the high-frequency (MHz) current can pass through the intracellular medium, but both the low (kHz) and high (MHz) currents can propagate through the extracellular ionic medium. A powerful understanding of this complex phenomenon can be obtained using equivalent electrical circuit models [4, 5], typically consisting of series and parallel combinations of resistors and capacitors (electrical component that can store charges) modeling for the variety of cell sizes and morphology.

Muscle Dielectric Properties and Anisotropy

An important characteristic of a skeletal muscle is the variation in impedance values when the electrical current flows along versus across the muscle fibers. This phenomenon, known as anisotropy, depends on the conductivity and permittivity properties of muscle. The anisotropy ratio measures the difference in the directional dependence of these properties. Under certain assumptions [6–11], the anisotropy ratio can be approximated simply as

$$\left(\frac{R_T}{R_L}\right)^2 \text{and} \left(\frac{X_T}{X_L}\right)^2$$

where $R_{L,T}$ and $X_{L,T}$ are the resistance and reactance components of the impedance measured with the four-electrode probe oriented in the longitudinal (along the muscle fibers) and transverse (across the muscle fibers) directions.

Measuring the dielectric properties has shown value, for example, in identifying various types of pathology in the muscle [12•] and differentiating type I from type II fibers [13]. In diseased muscle, such as in amyotrophic lateral sclerosis (ALS), the degeneration and loss of motor neurons associated with the disease impact the anisotropic nature of the dielectric properties of muscle [14]. Similarly, a recent study in mice indicated the loss of anisotropy in the dystrophic muscle that could serve as a surrogate indicator of disease progression in boys with the Duchenne muscular dystrophy [15].

Physiological Alterations in Diseased Muscle and Relationship with EIM

One of the challenges in EIM is relating observed alterations with the tissue's underlying cellular structure. We know that myopathies may be characterized by large numbers of inflammatory cells and edema, along with fat and connective tissue deposition. On the other hand, neurogenic diseases are associated predominantly with severe myofiber atrophy and fiber type grouping. Ultimately, these various characteristics will impact the impedance values in different ways. As an example, the peak reactance frequency in EIM, also known as characteristic frequency (*fc*), can be estimated using a physiological model consisting of the membrane capacitance C_m, the intracellular resistivity ρ_i, and the radius of the myofibers r as follows [16]

$$fc \approx \frac{1}{\pi r C_m \rho_i}.$$

Details aside, this concept suggests that cellular features can be expected to alter the measured impedance values in predictable, albeit complex, ways.

EIM in Neuromuscular Disorders: an Overview

EIM as a Primary Diagnostic

The simplest, most basic diagnostic problem is that of distinguishing healthy from diseased muscle without attempting to reach a specific neuromuscular diagnosis. A number of studies have shown that a muscle impacted by disease will have different EIM values than a healthy muscle; this has been demonstrated across populations of individuals most clearly in ALS [17]; spinal muscular atrophy (SMA) [18]; radiculopathy [19]; muscular dystrophies, including the

Duchenne muscular dystrophy (DMD) [20]; and facioscapulohumeral muscular dystrophy (FSHD) [21•]. Importantly, except in the case of radiculopathy [19] (as described below), in most of these cases, the diagnostic accuracy (i.e., sensitivity and specificity) of the test was not specifically sought, but rather simple group differences were demonstrated. In some of these, such as in DMD, there may be substantial overlap between healthy and diseased, especially between those with mild or early disease. Unsurprisingly, with more advanced disease, distinguishing between healthy and diseased subjects becomes simpler. Table 1 provides an overview of EIM's application across diseases.

In terms of differentiating between disease groups (rather than merely between healthy and diseased), the most basic problem that can be posed is that of distinguishing between primary neurogenic and primary myopathic diseases; a secondary question is whether these conditions can be differentiated from simple disuse change. From a theoretical perspective, differentiating among all three groups (ie., healthy, neurogenic, and myopathic) could be theoretically possible. One study in mice evaluated muscle tissue ex vivo from healthy animals and from three groups of animals each with a different condition: ALS, DMD (via the *mdx* mouse), and disuse atrophy (produced via hind limb suspension). It was demonstrated that the conductive properties, including their anisotropic characters, differentiated between the four states.

The one published human study that attempted disease differentiation between myopathy, neurogenic, and healthy evaluated multidirectional anisotropic measurements over the tibialis anterior alone [14]. That study demonstrated that compared to healthy subjects, there was a subtle loss of muscle anisotropy in the myopathy patients with an increase in anisotropy in the ALS patients. Again, there was no specific effort to identify the sensitivity or specificity of this diagnostic approach. More recently, a study sought to evaluate EIM's potential use as a means of discriminating ALS patients from those with mimicking disorders and from healthy individuals [Rutkove, unpublished results]. The test was effective at distinguishing healthy from ALS showing good accuracy (receiver operating characteristic (ROC) area under the curve = 0.80), but was of limited use in distinguishing ALS patients from disease mimics (ROC area under the curve = 0.62).

EIM for Assessing Disease Status, Progression, and the Effects of Therapy

Neurogenic Disorders

Amyotrophic Lateral Sclerosis The first case study of EIM identified progressive reduction in 50-kHz EIM phase values in the quadriceps of a man diagnosed with ALS prior to that muscle's being clinically affected [22]. The ability to track ALS progression was confirmed in two subsequent studies [23, 24]. The second of these showed that EIM had the potential for reducing the sample size of a clinical trial by more than 50% compared to standard clinical outcomes [24]. A third recently completed multicenter study utilized a dedicated EIM system and suggested a greater than fivefold improvement in sensitivity to progression (unpublished). More

Table 1 Overview of conditions in which EIM has been studied

	Human subjects	Animal models	Comments
Neurogenic disorders			
Amyotrophic lateral sclerosis	✓	✓	Identifies disease progression; correlates to standard functional electrophysiological measures
Spinal muscular atrophy	✓	✓	Identifies disease progression; correlates to standard functional electrophysiological measures
Radiculopathy	✓	✓	Can be used to assist with diagnosis with accuracy similar to EMG
Nerve injury	✓	✓	Can be used to track recovery
Myopathic disorders			
Duchenne muscular dystrophy	✓	✓	Sensitive to disease progression; correlates with functional measures
FSHD	✓	✓	Correlates to magnetic resonance imaging
Myotonic dystrophy	✓	✓	Value uncertain
Inflammatory myopathies	✓	✓	Correlates to functional alterations
Disuse/aging/trauma			
Sarcopenia	✓	✓	Differentiates sarcopenic versus healthy muscle; correlates to function in older individuals
Disuse	✓	✓	Sensitive to progression and recovery; changes are subtler than in primary neuromuscular disease
Muscle injury	✓	✓	Has been used to study traumatic and stretch injury

recently, EIM has been used to evaluate bulbar dysfunction in ALS patients via measurement of the tongue [25, 26]; this new application is especially relevant since there are few easily quantifiable bedside bulbar assessment tools available.

In parallel, research efforts on mouse and rat ALS models [27, 28] have allowed us to establish the relationship between EIM and standard electrophysiologic and functional measures [27, 29], including the compound motor unit action potential (CMAP) amplitude, the motor unit number estimate (MUNE) [30], and force [31].

Spinal Muscular Atrophy EIM has been shown to correlate to disease severity in milder forms of spinal muscular atrophy (types 2 and 3) [18]. Longitudinal studies have demonstrated that in healthy children, the 50-kHz phase gradually increases, likely due to myofiber growth and maturation, whereas in older children with untreated SMA, values remain relatively stable [32]. A multicenter study in infants with type 1 SMA showed EIM's ability to distinguish from healthy children at baseline [33•] and to detect differences in the EIM values with growth. Studies in animals confirmed EIM's sensitivity to treatment effect using antisense oligonucleotides in the SMNΔ7 mouse [34], similar to CMAP and MUNE; however, unlike those modalities, the data can be acquired without nerve stimulation [35].

Nerve Injury and Radiculopathy Studies have shown EIM usefulness to diagnose radiculopathy comparing measures on the affected side to those on the unaffected side [19], achieving a similar performance to that of needle electromyography (EMG) [19, 36]. Analogous results to those of CMAP and MUNE were found in rat models of sciatic crush [30], with decreased EIM phase values after injury that gradually return to baseline values over time. One study has also identified the potential for EIM to discriminate among different traumatic nerve injuries in the upper extremity [37].

Carpal Tunnel Syndrome Another potential application of EIM is the assessment of carpal tunnel syndrome (CTS), the most common compression neuropathy. A recent study showed that EIM values of affected abductor pollicis brevis in CTS patients are lower than those for healthy subjects, with no significant differences in unaffected muscles [38].

Other Neurogenic Disorders To our knowledge, EIM has not been employed in other motor or sensorimotor axonal polyneuropathies, probably because standard electrophysiological techniques can be used to easily identify them. Similarly, diseases of the neuromuscular junction, such as myasthenia gravis and the Lambert-Eaton myasthenic syndrome, have also not been studied.

Myopathic Conditions

Duchenne Muscular Dystrophy A study from our group has shown that EIM 50-kHz phase can distinguish boys with DMD from healthy children [39]. Further, clinical studies have found that EIM phase is sensitive to disease progression in both older (> 7 years) and younger (< 7 years) boys [40•], being also sensitive to treatment effect using steroids [40•]. In the *mdx* mouse model, EIM was found to have different values as compared to wild-type animals, with correlation of EIM values to connective tissue content [41]. This correlation was confirmed with the T2 signal on MRI [42]. EIM has also shown sensitivity to the impact of myostatin inhibition in both wild-type and *mdx* mice [43].

Facioscapulohumeral Muscular Dystrophy EIM reactance values in patients with FSHD were found to correlate with the degree of abnormality on MRI [21•]. The flexibility offered by EIM to evaluate affected muscles can be an especially useful tool in a disease with a highly variable presentation, as is the case with FSHD.

Myotonic Dystrophy To our knowledge, myotonic dystrophy has been studied in mouse models only. Data [unpublished] have suggested EIM's capability to detect both the presence of the disease and potentially the effect of therapy.

Inflammatory Myopathies Early work showed that EIM phase was reduced in individuals with inclusion body myositis (IBM), likely because of the presence of more resistive fat tissue in the quadricep muscle [22]. EIM phase correlated as well with the clinical status of patients with polymyositis and dermatomyositis [44].

Critical Illness Myopathy Technical challenges aside, EIM could be a suitable and convenient tool for the evaluation of patients in the intensive care unit with critical illness myopathy. Further research is warranted to assess this potential application.

Sarcopenia, Disuse, and Cachexia

Applying EIM to these groups of disorders could complement standard electrodiagnostic outcomes, which are expected to be normal or virtually normal using needle EMG [45]. For example, diseases such as sarcopenia have an associated concomitant increase in connective tissue and to some extent fat in the muscle. While these changes are not be detected by needle EMG, they will substantially impact the EIM data.

Sarcopenia A cross-sectional study, involving 100 subjects across the age spectrum, revealed that EIM phase values decreased overall with increasing age [46]. Regardless of the

sex, a non-linear relationship was found between EIM 50-kHz phase in the tibialis anterior and gastrocnemius. Further, EIM reduction with age was most pronounced in both sexes for people over 60 years. Longitudinal measurements on a small set of four patients over 75 years of age without any superimposed neuromuscular condition showed a decline over time.

Reductions in the 50-kHz reactance were found to be more evident in a subsequent cross-sectional study measuring both lower and upper extremities, being more obvious in men than women [47]. Both studies were in good agreement with previous work showing that other markers of sarcopenia were more marked in men than women [48], where changes in lower extremities were more pronounced than those in the upper extremities [49].

More recently, EIM values measured in the lower extremities of older adults have been found to correlate with function measures such as force (unpublished). The possibility of carrying out EIM measures at home with minimal training makes EIM a potentially valuable tool for self-assessment and evaluating the effects of interventions.

Disuse Atrophy In patients that had suffered an ankle fracture, after the cast was removed and the patient became fully weight-bearing, EIM phase values gradually increased with recovery [50]. Consistent observations were found in hind limb-suspended rats [51]. After 2 weeks of suspension, EIM phase values dropped by 33%. Two weeks after having been released from suspension, the animals had partially recovered being 19.5% lower than baseline. Disuse atrophy in mice under the effect of microgravity aboard the NASA STS-135 space shuttle mission in 2011 also showed reductions in EIM phase values. In this study, however, phase values were on average 43% lower than the control group [52]. Future work should investigate changes occurring in cachexia, a disease condition with similar pathology as disuse atrophy and sarcopenia.

Traumatic Injury of the Muscle

Traumatic injury to the muscle has been also studied longitudinally in elite soccer players and confirmed with ultrasound and magnetic resonance imaging [53, 54]. Muscle injuries included hematoma due to direct trauma and muscle tears of different severities in the lower extremity muscles. The EIM values on the injured side decreased with injury and gradually recovered over time as the injury resolved. These findings support the idea of using EIM as an easily applied biomarker to predict the return-to-play time [55].

Muscle injury induced by eccentric contractions has been studied in *mdx* and healthy mouse quadricep muscles [56]. Significant reductions in phase values were observed over a period of 48 h caused by the presence of edema as confirmed with increased T2 signal on MRI. Modeling the data revealed major changes occurring to the characteristic frequency in both WT and *mdx* mice, likely caused by the disruption of the sarcolemma and t-tubule system after injury.

Current and Future Directions

Further theoretical, engineering, and clinical works need to be completed before EIM can be expected to reach its full potential. Moreover, the specific requirements for its successful implementation in each of these disorders also need to be better defined.

1. *Clinical trials*. Given EIM's high reproducibility [57] and the fact that it requires minimal patient cooperation, EIM may offer a reliable and objective alternative to standard functional outcome measures. For example, EIM can be potentially useful in DMD trials where the standard 6-min walk test depends on patients' mood and effort. However, there still remains work to be completed, depending on the specific clinical application, including improved measurement/analysis approaches. For example, in one ongoing study in ALS [58], a smaller EIM probe is being used to evaluate hand muscles—a body region which is markedly impacted by the disease—but that has to date remained mostly unstudied with EIM.
2. *Relationship to pathology*. Multifrequency impedance modeling is a potentially powerful approach for analyzing the physiological alterations associated with a diseased muscle beyond solely detecting changes in single-frequency EIM values. For example, as noted earlier, modeling the characteristic frequency could provide insights into changes occurring on the myofibers' radii [41]. Another example is that such multifrequency modeling can also provide pathological information into intracellular characteristics such as

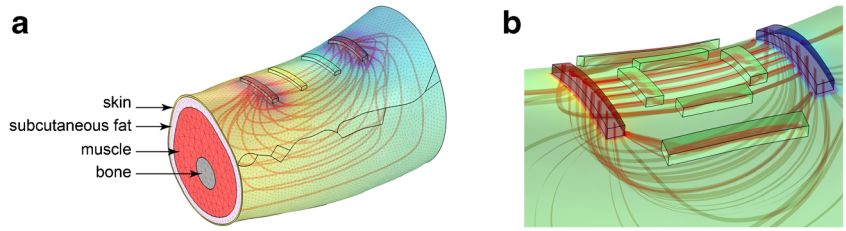

Fig. 1 Illustrative model simulation that incorporates a realistic limb shape (**a**) and detail of the current density near the electrodes (**b**)

mitochondrial content in type 1 and type 2 myofibers [13]. Ongoing animal research is focused on establishing multifrequency impedance model relationship to muscle compositional and structural information including myofibers' radii, presence of interstitial fat tissue, glycogen storage abnormalities, and myofiber disorganization.

3. *Finite element model simulations.* Whereas the term "myography" in EIM's name implies that the technique assesses muscle condition alone, in practice, surface EIM values will have contributions from skin and subcutaneous fat. Thus, additional research is required to reduce the effect of non-muscle tissue by improving the electrode characteristics (e.g., size, spacing between electrodes, arrangement, and material) to optimize EIM's sensitivity to muscle status. As of today, however, much of the work has used linear electrode arrangements that were not optimized [59]. Combining theory with finite element model simulations, in which estimations of the expected surface impedance values can be estimated based on known tissue properties (see an example in Fig. 1), is required to fully exploit the value of EIM in the assessment of muscle condition across the disease and injury spectrum [60].
4. **Impedance during contraction**. Remarkably, the basic concept of using impedance to measure muscle contraction dates back nearly 100 years but has met with inconsistent results [61–64]. This idea is attractive since it could offer insights into the mechanics of muscle contraction in health and disease, one area in which standard electrophysiologic techniques are relatively poor. Current standard approaches, such as needle EMG or nerve conduction studies, only measure up to the point of muscle fiber depolarization, ignoring entirely the contractile mechanism itself. While technically challenging, recent animal studies have suggested that impedance alterations in muscle contraction are disease-dependent [65].

Conclusion

Based on the large number of recent or ongoing studies employing EIM, it is clear that the technique may have wide value in the assessment of neuromuscular disorders and muscle health broadly. Although the underlying principles of bioimpedance theory may be unfamiliar to most medical professionals, we believe that as people become increasingly comfortable with the terminology and concepts of EIM, given the technique's remarkable simplicity of application, it will gradually be accepted into a neurologist's standard toolbox of diagnostic tests.

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Compliance with Ethical Standards

Conflict of Interest Benjamin Sanchez has received personal fees from ImpediMed, Inc., and Maxim Integrated, Inc. In addition, Dr. Sanchez has a patent 2015123603 A1 licensed to Beth Israel Deaconess Medical Center.

Seward B. Rutkove has equity in and serves a consultant and scientific advisor to Skulpt/Myolex, Inc., a company that designs impedance devices for clinical and research uses; he is also a member of the company's board of directors. The company also has an option to license patented impedance technology of which Dr. Rutkove is named as an inventor. In addition, Dr. Rutkove has a patent US Patent #9,014,797, electrical impedance myography licensed to Skulpt, Inc.; a patent US Patent #8,892,198, devices and methods for evaluating tissue licensed to Skulpt, Inc.; a patent US patent application: a hand-held device for electrical impedance myography pending; and a patent PCT/US2015/015961, electrical impedance myography pending.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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